08 Text in Space, Truth in the Schema: A FOSS Neuro-Symbolic Middleware for Querying Roman Archaeology without Hallucinations (IN-ROME Project)

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The IN-ROME project aims at a holistic description of imperial Rome and its suburbium (within a 13 km radius), integrating a massive heterogeneous corpus of archaeological, epigraphic, and spatial evidence into a relational database. The spatial framework bridges the SITAR Rome GIS, the Carta Archeologica/Lazio Antico (Sapienza, Prof. P. Carafa), and over 4,000 sites surveyed by the research team. Onto this landscape are mapped instrumenta domestica (over 4,400 brick stamps and lead pipes from CIL XV, digitized via OCR and parsing and geolocalized) and a prosopographical network of over 5,200 individuals with attested social relations. Supported by an EDR gazetteer of over 10,000 georeferenced toponyms, inscriptions become true "text in the space".

Querying this complex data requires an adequate engine: traditional systems fail due to literal matching (text = text), while generative LLMs risk hallucinations. We present an open-source (FLOSS) neuro-symbolic middleware that connects an on-premise open-weight LLM, served through Ollama, to a relational and geospatial database (PostgreSQL/PostGIS) through a domain-specific semantic layer combining controlled vocabularies, typed relations, evidence-aware inference rules, external Linked Open Data (LOD) alignments, and explicit mappings to the underlying database schema. The core is a declarative Semantic Layer formalizing concepts, relationships, vocabularies, and inference rules, anchored to Linked Open Data standards (Pleiades, CIDOC-CRM/EAGLE, Getty AAT). The paper tackles the hallucination paradox: when parsing texts, the LLM may hallucinate; however, output is not buried in opaque "black boxes", but in open YAML files validated by archaeologists (Human-in-the-Loop). At run-time, the validated layer binds the LLM to compile deterministic SQL/PostGIS queries, eliminating hallucinations.

Finally, the new framework is engineered for community reuse: by decoupling domain semantics from physical schemas, other Roman research projects can adopt the Semantic Layer via a mapping file (Schema Adapter), inheriting this conceptual engine and advancing Open Science.